

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 5

For the tutorial in Week 6 on 22 February, let us discuss

- Ex. 8.1.9, 8.1.11, 8.2.2(a)–(e), 8.2.5, 8.2.6, additional exercises.

Additional Exercise on tiling. You need to tile a $n \times 1$ hallway with unlimited supply of 1×1 red tiles, 2×1 red tiles, and 2×1 blue tiles. Write down the recurrence formula and initial conditions for the number of ways to tile:

- (a) an $n \times 1$ hallway for which you are allowed to use all types of tiles.
- (b) an $n \times 1$ hallway for which you are allowed to use all types of tiles, but no two red tiles are consecutive.

Solution:

- (a) Let $f(n)$ be the number of ways to tile an $n \times 1$ hallway with no restrictions on the type of tiles.

Case 1: The last tile is a 1×1 red tile. There are $f(n-1)$ ways since there are no restrictions on the first $(n-1) \times 1$ hallway.

Case 2: The last tile is a 2×1 red tile. There are $f(n-2)$ ways since there are no restrictions on the first $(n-2) \times 1$ hallway.

Case 3: The last tile is a 2×1 blue tile. Similarly, there are $f(n-2)$ ways.

Therefore the number of possible ways $f(n)$ to tile an $n \times 1$ hallway satisfies the recurrence

$$f(n) = f(n-1) + 2f(n-2), \quad \text{for } n \geq 3.$$

The initial conditions are $f(1) = 1$ and $f(2) = 3$.

- (b) Let $g(n)$ be the number of ways to tile an $n \times 1$ hallway with the restriction that no two red tiles are consecutive.

Case 1: The last tile is a 1×1 red tile. Then the penultimate tile must be a 2×1 blue tile. For the remaining $n-3$ tiles, they just need to satisfy the requirement that there are no consecutive red tiles. There are $g(n-3)$ possible ways.

Case 2: The last tile is a 2×1 red tile. Similarly, the penultimate tile must be a 2×1 blue tile and there are $g(n-4)$ possible ways.

Case 3: The last tile is a 2×1 blue tile. By the same reason, there are $g(n-2)$ possible ways.

Therefore, the recurrence relation is

$$g(n) = g(n-2) + g(n-3) + g(n-4), \quad \text{for } n \geq 5.$$

The initial conditions are $g(1) = 1, g(2) = 2, g(3) = 2$, and $g(4) = 4$.

Ex. 8.1.9. A ternary string is a string containing only 0's, 1's, and 2's.

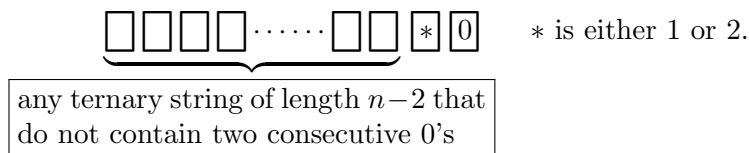
- (a) Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's.
- (b) What are the initial conditions ?
- (c) Find the answer when $n = 4$.

Solution:

- (a) Let a_n be the number of ternary strings of length n that do not contain two consecutive 0's. Then the recurrence equation is $a_n = 2a_{n-2} + 2a_{n-1}$.

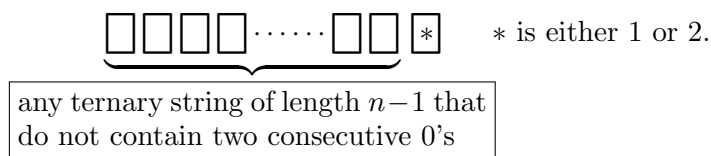
Consider the last bit of the string.

Case 1a & 1b: The last bit is a 0. Then the penultimate bit must be either a 1 or a 2. In each case, the first $n-2$ bits are any ternary string of length $n-2$ that do not contain two consecutive 0's, and there are a_{n-2} such strings.



Case 2: The last bit is 1. Then the first $n-1$ bits are any ternary string of length $n-1$ that do not contain any two consecutive 0's. There are a_{n-1} such strings for each case.

Case 3: The last bit is 2. This is similar to Case 2, and there are a_{n-1} such strings.



Therefore, $a_n = 2a_{n-1} + 2a_{n-2}$, for $n \geq 3$.

- (b) $a_1 = 3, a_2 = 8$.
 (c) $a_3 = 2a_1 + 2a_2 = 22$, then $a_4 = 2a_2 + 2a_3 = 60$.

Ex. 8.1.11.

- (a) Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's.
 (b) What are the initial conditions?
 (c) Find the answer when $n = 4$.

Solution:

Let a_n be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's.

- (a) Define $a_{0,n}$ to be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's and ends with 0. Define $a_{1,n}$ and $a_{2,n}$ similarly. Then clearly

$$a_n = a_{0,n} + a_{1,n} + a_{2,n}.$$

Note that when the last bit of such a ternary string of length n is 0, then the penultimate bit has to be either 1 or 2. Therefore,

$$a_{0,n} = a_{1,n-1} + a_{2,n-1}.$$

Likewise, for the case when the last bit of such a ternary string of length n is 1, we have

$$a_{1,n} = a_{0,n-1} + a_{2,n-1}.$$

When the last bit of such a ternary string of length n is 2, the first $n-1$ bits can be any ternary string of length $n-1$ not containing two consecutive 0's or two consecutive 1's. Therefore,

$$a_{2,n} = a_{n-1} = a_{0,n-1} + a_{1,n-1} + a_{2,n-1}.$$

Putting all three equations together, we see that

$$\begin{aligned}
 a_n &= a_{0,n} + a_{1,n} + a_{2,n} \\
 &= (a_{1,n-1} + a_{2,n-1}) + (a_{1,n-1} + a_{2,n-1}) + (a_{0,n-1} + a_{1,n-1} + a_{2,n-1}) \\
 &= 2(a_{0,n-1} + a_{1,n-1} + a_{2,n-1}) + a_{2,n-1} \\
 &= 2a_{n-1} + a_{n-2}, \quad \text{valid for } n \geq 3.
 \end{aligned}$$

(b) $a_1 = 3, a_2 = 7$.

(c) $a_3 = 2a_2 + a_1 = 17$, then $a_4 = 2a_3 + a_2 = 41$.

Ex. 8.2.2 (a)–(c). Solve the following homogeneous recurrence relations.

(a) $a_n = 2a_{n-1}$ for $n \geq 1, a_0 = 3$.

(b) $a_n = a_{n-1}$ for $n \geq 1, a_0 = 2$.

(c) $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2, a_0 = 1, a_1 = 0$.

Solution:

(a) This is a homogenous recurrence relation with degree 1. The characteristic equation is $x = 2$, i.e., root $r = 2$ and $a_n = h \cdot 2^n$. Substitute $3 = a_0 = h$, hence $a_n = 3 \cdot 2^n$.

(b) This is a homogenous recurrence relation with degree 1. The characteristic equation is $x = 1$, i.e., root $r = 1$ and $a_n = h \cdot 1^n = h$. Substitute $2 = a_0 = h$, hence $a_n = 2$.
Alternatively, one can also see that $a_n = a_{n-1} = a_{n-2} = \dots = a_0 = 2$.

(c) This is a homogenous recurrence relation with degree 2. The characteristic equation is $x^2 = 5x - 6 \implies x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, i.e., there are two distinct roots $r_1 = 2$ and $r_2 = 3$, hence

$$a_n = h_1 \cdot 2^n + h_2 \cdot 3^n.$$

From the initial conditions, we see that

$$1 = a_0 = h_1 + h_2,$$

$$0 = a_1 = h_1 \cdot 2 + h_2 \cdot 3,$$

solving which results in $h_1 = 3, h_2 = -2$. Hence $a_n = 3 \cdot 2^n - 2 \cdot 3^n$.

Ex. 8.2.2 (d), (e). Solve the following homogeneous recurrence relations.

(d) $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2, a_0 = 6, a_1 = 8$.

(e) $a_n = -4a_{n-1} - 4a_{n-2}$ for $n \geq 2, a_0 = 0, a_1 = 1$.

Solution:

(d) This is a homogenous recurrence relation with degree 2. The characteristic equation is $x^2 = 4x - 4 \implies x^2 - 4x + 4 = (x - 2)^2 = 0$, i.e., there is only one root $r_0 = 2$, hence

$$a_n = h_1 \cdot 2^n + h_2 \cdot n \cdot 2^n.$$

From the initial conditions, we obtain

$$6 = a_0 = h_1,$$

$$8 = a_1 = h_1 \cdot 2 + h_2 \cdot 1 \cdot 2,$$

solving which results in $h_1 = 6, h_2 = -2$. Hence $a_n = 6 \cdot 2^n - n \cdot 2^{n+1}$.

- (e) This is a homogenous recurrence relation with degree 2. The characteristic equation is $x^2 = -4x - 4 \implies x^2 + 4x + 4 = (x + 2)^2 = 0$, i.e., there is only one root $r_0 = -2$, hence

$$a_n = h_1 \cdot (-2)^n + h_2 \cdot n \cdot (-2)^n.$$

From the initial conditions, we obtain

$$0 = a_0 = h_1 + h_2 \cdot 0,$$

$$1 = a_1 = h_1 \cdot (-2) + h_2 \cdot 1 \cdot (-2),$$

solving which results in $h_1 = 0, h_2 = (-2)^{-1}$. Hence $a_n = n \cdot (-2)^{n-1}$.

Ex. 8.2.5. Let a_n be the number of ways to tile a $2 \times n$ rectangular board using 1×2 and 2×2 pieces.

- Calculate the values a_1 and a_2 .
- Find a recurrence relation for the sequence (a_n) .
- Solve the recurrence relation in part (b).

Solution:

- Initial conditions are $a_1 = 1, a_2 = 3$.
- Consider the last column of the board, it can be filled either by
 - a 2×2 piece (which has a_{n-2} ways for the remaining), or
 - a 1×2 piece vertically (which has a_{n-1} ways for the remaining), or
 - two 1×2 piece horizontally (which has a_{n-2} ways for the remaining).

Hence the recurrence relation is $a_n = a_{n-1} + 2a_{n-2}$, for $n \geq 3$.

- Its characteristic equation is $r^2 - r - 2 = 0$ with roots $r_1 = 2$ and $r_2 = -1$. Since the two roots are distinct, we have

$$a_n = h_1 \cdot 2^n + h_2 \cdot (-1)^n.$$

By the initial conditions, $a_1 = 1 = 2h_1 - h_2$ and $a_2 = 3 = 4h_1 + h_2$, which implies that $h_1 = 2/3$ and $h_2 = 1/3$. So the final answer is

$$a_n = \frac{2}{3} \cdot 2^n + \frac{1}{3} \cdot (-1)^n.$$

Ex. 8.2.6. A model for the number of lobsters caught per year is based on the assumption that the number of lobsters caught in a year is the average of the number caught in the two previous years.

- Find a recurrence relation for the sequence (L_n) , where L_n is the number of lobsters caught in year n .
- Solve L_n if 100,000 lobsters were caught in year 1 and 300,000 lobsters were caught in year 2.

Solution:

- The recurrence relation is, by the definition of average, $L_n = (1/2)L_{n-1} + (1/2)L_{n-2}$.

- (b) The characteristic equation is $r^2 - (1/2)r - 1/2 = 0$, which gives two distinct roots, $r_1 = -1/2$ and $r_2 = 1$. Therefore the general solution is

$$L_n = h_1(-1/2)^n + h_2.$$

From the initial conditions $L_1 = 100000$ and $L_2 = 300000$, we obtain

$$100000 = (-1/2)h_1 + h_2,$$

$$300000 = (1/4)h_1 + h_2.$$

Solving these yields $h_1 = 800000/3$ and $h_2 = 700000/3$. Therefore the answer is

$$L_n = (800000/3)(-1/2)^n + (700000/3).$$

Additional Exercise. Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1},$$

$$b_n = a_{n-1} + 2b_{n-1},$$

with $a_0 = 1$ and $b_0 = 2$.

Solution: From $b_n = a_{n-1} + 2b_{n-1}$, rearrange to get

$$a_{n-1} = b_n - 2b_{n-1}. \quad (1)$$

Replacing n by $n+1$ in (1), we obtain

$$a_n = b_{n+1} - 2b_n. \quad (2)$$

Substituting (1) and (2) into the equation $a_n = 3a_{n-1} + 2b_{n-1}$, we find that

$$(b_{n+1} - 2b_n) = 3(b_n - 2b_{n-1}) + 2b_{n-1}.$$

Upon simplifying and rearranging, we arrive at

$$\begin{aligned} b_{n+1} &= 3b_n - 6b_{n-1} + 2b_{n-1} + 2b_n \\ &= 5b_n - 4b_{n-1}. \end{aligned}$$

This is a homogeneous recurrence of degree 2 with characteristics equation $r^2 = 5r - 4$ with two roots $r_1 = 4$ and $r_2 = 1$. Hence the solution $b_n = h_1 \cdot 4^n + h_2$, with initial conditions

$$2 = b_0 = h_1 + h_2,$$

$$5 = b_1 = 4h_1 + h_2. \quad (\text{Note that } b_1 = a_0 + 2b_0.)$$

Hence $h_1 = 1$ and $h_2 = 1$. Thus $b_n = 4^n + 1$. From (2),

$$\begin{aligned} a_n &= b_{n+1} - 2b_n \\ &= 4^{n+1} + 1 - 2(4^n + 1) \\ &= 2 \cdot 4^n - 1. \end{aligned}$$

Hints.

Additional Exercise on tiling.

- (a) Instead of considering the last bit, consider the last tile, is it a 1×1 red tile, or 2×1 red tile, or 2×1 blue tile?
- (b) If the last tile is a red tile, then the penultimate tile must be a blue tile.

Ex. 8.1.9. Similar to 8.1.10(a), consider the last bit, or the last two bits if necessary.

Ex 8.1.11. Let a_n be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's. Define $a_{0,n}$ to be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's and **ends with 0**. Define $a_{1,n}$ and $a_{2,n}$ similarly. Then clearly

$$a_n = a_{0,n} + a_{1,n} + a_{2,n}.$$

Express $a_{0,n}$ in terms of $a_{0,n-1}$, $a_{1,n-1}$, and $a_{2,n-1}$. Do likewise for $a_{1,n}$, and $a_{2,n}$.

Ex 8.2.5 As usual, consider the last column of the board. Note that a 1×2 piece can be placed horizontally or vertically.